Technology

A. React

B. Vue.js

C. Angular

D. Express (Node.js)

E. Laravel (PHP)

F. ASP.NET (C#)

G. Django (Python)

Select one of the following options:

Option	Frontend Framework	Backend Framework
1	**React**	**Express (Node.js)**
2	**Vue.js**	**Express (Node.js)**
3	**Angular**	**Express (Node.js)**
4	**Laravel (PHP)**	— (Full-stack framework)
5	**ASP.NET (C#)**	— (Full-stack framework)
6	**Django (Python)**	— (Full-stack framework)

1. A group may consist of a maximum of 5 students.
2. Every group must register their team members and select technology option until October 23, 2025.
3. No more than two groups are allowed to choose the same technology option from the list above.
4. Technology options will be allocated on a first-come, first-served basis.
5. Create a GitLab repository for your project.

List of Projects for DBMS(CSE 3rd Year 2025)

1. Hotel Reservation System

   - ✓ Add new guest details (name, contact info, address, etc.).
   - ✓ Update or delete guest information.
   - ✓ Add new rooms to the system (room number, type, availability, price, etc.).
   - ✓ Update room details (e.g., change price or availability).
   - ✓ Book a room for a guest (check-in/check-out dates, number of guests).
   - ✓ Modify existing reservations (e.g., change dates or room type).
   - ✓ Cancel reservations.
   - ✓ Role-based access (e.g., admin, receptionist, guest).
   - ✓ Login and logout functionality.
   - ✓ Generate daily, weekly, or monthly reports

2. Inventory Management System

   - ✓ Role-based access (e.g., admin, inventory manager, or viewer).
   - ✓ Login/logout functionality.
   - ✓ Add new products (name, category, price, quantity, supplier, etc.).
   - ✓ Update product details (e.g., price, description, or quantity).
   - ✓ View product details with category or stock levels.
   - ✓ Delete products from the inventory.
   - ✓ Add new suppliers (name, contact details, address, etc.).
   - ✓ Update or delete supplier details.
   - ✓ Record incoming stock orders (purchase orders).
   - ✓ rack outgoing stock for sales or deliveries.
   - ✓ Organize products by categories (e.g., electronics, clothing, food).
   - ✓ Add, update, or delete categories.
   - ✓ Generate stock reports (e.g., low-stock products, most sold items).

- ✓ Handle stock adjustments for damaged, expired, or returned goods.
- ✓ Generate barcodes or QR codes for products.
- ✓ Scan codes to retrieve product details or update stock.

3. Hospital Management System

- ✓ Role-based access (e.g., admin, doctor, receptionist, pharmacist).
- ✓ Login/logout functionality.
- ✓ Add new patient details (name, age, gender, contact info, address, medical history, etc.).
- ✓ Update or delete patient records.
- ✓ Search and retrieve patient details.
- ✓ Add, update, or delete doctor details (name, specialization, availability, contact info).
- ✓ Assign doctors to patients.
- ✓ View doctor schedules and appointments.
- ✓ Schedule new appointments (patient ID, doctor ID, date, time, purpose).
- ✓ Update or cancel appointments.
- ✓ View upcoming appointments for patients and doctors.
- ✓ Maintain patient medical history and link records to patients and doctors.
- ✓ Add and manage staff details (nurses, receptionists, technicians, etc.).
- ✓ Assign roles and responsibilities to staff members.

4. Library Management System

- ✓ Role-based access (e.g., admin, librarian, member).
- ✓ Login and logout functionality. Add new books and Update book details (e.g., title, author).
- ✓ Remove books from the system.
- ✓ Search and filter books by title, author

- ✓ Add new library members (name, contact info)
- ✓ Update or delete member details.
- ✓ Issue books to members (book ID, member ID, issue date, due date).
- ✓ Record book returns and track currently borrowed books.
- ✓ Organize books by categories

5. Restaurant Management System

- ✓ Add, update, or delete menu items (name, category, price, description, availability).
- ✓ Place new orders (table number, customer name, menu items, quantity).
- ✓ Update or cancel existing orders.
- ✓ Track order status (e.g., "Pending," "Preparing," "Served").
- ✓ Add and manage tables (table number, capacity, availability).
- ✓ Assign tables to customers and Mark tables as "Occupied" or "Available."
- ✓ Add and manage staff details (name, role, contact info, shifts, salary).
- ✓ Track staff schedules and responsibilities (e.g., waiter, chef, cashier).

6. HR Management System

- ✓ Add, update, and delete employee details (name, contact info, department, designation, salary, etc.).
- ✓ Maintain employee profiles (personal details, qualifications, experience).
- ✓ View employee records based on filters (e.g., department, job title).
- ✓ Track job openings (job title, department, requirements, status).
- ✓ Manage applicant data (name, contact, resume, application status).

- ✓ Stores department details (DepartmentID, Name, Description) and

7. Online Rental System
   - ✓ Browse rental listings
   - ✓ Search/filter by category, price, location
   - ✓ View item details (images, description, price)
   - ✓ Book rental items (select dates, view total cost)
   - ✓ View/Cancel bookings
   - ✓ Add/edit/delete rental items
   - ✓ Manage booking requests (approve/decline)
   - ✓ Register account and manage all users (suspend/activate)
8. Tourist Management System
   - ✓ Create role such as Tourist (Guest/Registered), Service Provider (Business), Administrator
   - ✓ Search/filter by location, category, rating
   - ✓ View detailed listings (photos, prices, hours)
   - ✓ Add/update services (tours, hotels, activities)
   - ✓ Receive, confirm, or decline booking requests and View a dashboard of all bookings and their statuses.
9. School Management System
   - ✓ Manage academic years, class/batch creation, and subject allocation.
   - ✓ Create, update, and deactivate profiles for students, teachers, and staff.
   - ✓ Assign system roles and access levels to all users.
   - ✓ View system-wide reports, dashboards, and audit logs.
   - ✓ View personal timetable, assigned subjects, and teachers.
10. Online Food Ordering System

- ✓ Browse restaurants and menus
- ✓ Search/filter by price, ratings
- ✓ Place orders for delivery/pickup
- ✓ View order history
- ✓ Rate and review restaurants/orders
- ✓ Manage restaurant profile and menu
- ✓ Receive and manage orders
- ✓ Update order status (accepted, preparing, ready, etc.)
- ✓ View assigned delivery orders
- ✓ Accept/reject delivery requests
- ✓ Manage all users (customers, restaurants, drivers)
- ✓ generate report

11. Blood Bank Management System

- Donor registration and profile management
- ✓ View donation history and blood test results
- ✓ Blood request registration
- ✓ Search for specific blood types availability
- ✓ Generate blood compatibility reports
- ✓ User management (donors, staff, hospitals)
- ✓ Generate blood availability reports

12. Pharmacy Management System

- ✓ User registration and profile management
- ✓ Search medicines by name/type/condition
- ✓ Stock alert and expiry date tracking
- ✓ Add/update/delete medicine records
- ✓ User management (customers, staff, suppliers)
- ✓ Sales reports and Profit/loss calculation
- ✓ Barcode/QR code scanning
- ✓ Expiry date alerts